Sign up to discover human stories that deepen your understanding of the world.
Free for a limited time, unlimited access to hundreds of thousands of high-quality stories. Explore a vast collection of writing on topics you care about.
Sponsored by Medium

GraphRAG With Neo4j Building And LangChain | Building Streamlit UI To Consume Backend API

Vinod Rajendran · Follow
Published in AI Advances · Sep 2, 2024

Get app Try for $5/month

Welcome back, fellow developers! Today we're diving into another exciting chapter of our Neo4j GraphRAG from scratch series. In this tutorial, we'll explore how to build an interactive frontend using Streamlit that seamlessly integrates with the asynchronous API we developed in our previous article.

This hands-on guide will walk you through creating a user-friendly interface that harnesses the power of our backend API.

What You'll Learn In This Guide:

1. Create a dynamic user interface with Streamlit (Note: We'll focus on advanced implementation rather than basic Streamlit concepts)
2. Integrate our FastAPI backend seamlessly with the Streamlit frontend

Let's get started!

Image by Code With Prince

Directory Setup

Before we dive into coding, let's ensure our project structure is properly organized. Here's the directory structure you'll need for the Streamlit application. If you encounter any setup challenges, refer to the GitHub repository for detailed guidance.

The Streamlit application will be housed within your `chatbot_backend/services/ui` folder. Navigate to this directory and execute the following commands:

```
$ mkdir streamlit_app
$ cd streamlit_app
$ touch app.py app_logic.py config.py utils.py app.py
```

```
$ tree .
.
├── app.py
├── app_logic.py
├── config.py
└── utils.py

0 directories, 4 files
```

Image by Code With Prince

```
$ tree -I "__pycache__" .
.
├── chatbot_backend
│   ├── .dockerignore
│   ├── .env
│   ├── .env.example
│   ├── .gitignore
│   ├── .pytest_cache
│   │   ├── .gitignore
│   │   ├── CACHEDIR.TAG
│   │   ├── README
│   │   └── v
│   │       ├── cache
│   │       │   ├── lastfailed
│   │       │   ├── nodeids
│   │       │   └── stepwise
│   │       ├── .gitignore
│   │       ├── CACHEDIR.TAG
│   │       ├── README
│   │       └── v
│   │       ├── cache
│   │       │   ├── lastfailed
│   │       │   ├── nodeids
│   │       │   └── stepwise
│   │       ├── .gitignore
│   │       ├── CACHEDIR.TAG
```

Happy coding! And see you next time, the world keeps spinning.

GitHub - Princekrampah/neo4j_graphrag_tutorial

Contribute to Princekrampah/neo4j_graphrag_tutorial development by creating an account on GitHub.

github.com

LIlm | Neo4j | Chatbots For Business | AI | Graphrag

113

Published in AI Advances

61K followers · Last published 1 hour ago

Democratizing access to artificial intelligence

Follow

Written by Prince Krampah

1.2K followers · 24 following

Hello there , am Prince a full-stack web developer, data science enthusiast, lover of Python Programming with a deep interest in deep learning, computer vision

Follow

No responses yet

Bhalchandra Madhekar

What are your thoughts?

More from Prince Krampah and AI Advances

Prince Krampah

Multi Agent Architecture In Multi-agent Systems | Multi-agent...

Hello guys, welcome back to another exciting article on working with multi-agent AI...

Apr 21, 2025 · 21

In AI Advances by Nov Tech

OpenAI Just Dropped Three Bombs, and One of Them Might...

While everyone's distracted by leadership drama, OpenAI quietly launched features th...

Jan 28 · 1.92K · 88

In AI Advances by Nov Tech

$830 Billion Vanished in One Week—Because This AI Found 500...

Claude Opus 4.6 discovered zero-day vulnerabilities autonomously, and Wall Stree...

Feb 8 · 2.4K · 51

In Python in Plain English by Prince Krampah

Open-Source S3 Alternatives| Introduction To MinIO

I have recently been building a backend application, we reached a point where we...

Sep 15, 2025 · 59

See all from Prince Krampah | See all from AI Advances

Recommended from Medium

In Level Up Coding by Fareed Khan

Building a Scalable, Production-Grade Agentic RAG Pipeline

Autoscaling, Evaluation, AI Compute Workflows and more

Dec 31, 2025 · 1.5K · 17

Joe Njenga

I Tried New Claude Code Ollama Workflow (It's Wild & Free)

Claude Code now works with Ollama, which takes the game to the next level for...

Jan 19 · 1.6K · 23

In Data Science Collective by Marina Wyss

AI Agents: Complete Course

From beginner to intermediate to production.

Dec 6, 2025 · 3.8K · 131

In Generative AI by Adham Khaled

Stanford Just Killed Prompt Engineering With 8 Words (And I...

ChatGPT keeps giving you the same boring response? This new technique unlocks 2×...

Oct 19, 2025 · 24K · 637

In Towards AI by Divy Yadav

9 RAG Architectures Every AI Developer Must Know: A Complet...

Architectures beyond Naive Rag to build reliable production AI Systems

Dec 19, 2025 · 1.7K · 35

Agent Native

Local LLMs That Can Replace Claude Code

Small team of engineers can easily burn >$2K/mo on Anthropic's Claude Code...

Jan 20 · 1.4K · 40

See more recommendations